4(a)	The principle of superposition states that when two or more travelling waves of the same type meet at a point in space, the resultant displacement at that point is the vector sum of the displacements due to each of the waves at that point.
4(b)(i)	For coherent waves, the frequency is the same. The period of both waves is equal to 44 units of t. From fig 4.2, the closest peaks (or zero crossings) of the two waves differ by 7 units of t. $\text{phase difference} = \frac{7}{44} \times 360 = 57.3^{\circ}$
4(b)(ii)	At bright fringe, the phase difference of wave A and B is 0 or multiple of 2π. The intensity of the bright fringe is the sum of the amplitudes of wave A and B. Using the values in Fig 4.2, the estimated bright fringe intensity closest to P $= 3.4 + 0.6 = 4.0$

	At dark fringe, the phase difference of wave A and B is odd multiples of π. Since the amplitudes of wave A and B are not the same, the intensity of the dark fringe is not zero. Using the values in Fig 4.2, the estimated dark fringe intensity closest to P $= 3.4 - 0.6 = 2.8$ $\frac{\text{intensity of light at the dark fringe}}{\text{intensity of light at the bright fringe}} = \frac{(2.8)^2}{(4.0)^2} = 0.49$
5(a)	By conservation of energy